





**First Year
Interconnect**

Sign Up to Be a Mentor

Welcome! Please sign up to join the **First Year Interconnect (FYI) Mentorship Program** at University of Victoria.

☐ I have read and agree to the [Privacy Policy](#) and to the [Terms of Use](#)

☐ I am a citizen of the ☐ [European Economic Area](#) ⓘ

 **Continue With NetLink ID**

Not sure what's going on? [Learn More](#)

Become a First Year Interconnect (FYI) Mentor!

Are you an upper-year UVic student interested in supporting a first-year student as they transition to university? The First-Year Interconnect (FYI) peer mentorship program is looking for additional upper-year students to become mentors for the 2026–27 academic year. As an FYI mentor, you'll have the opportunity to share your experiences, help a new student navigate their transition to UVic, connect them with campus resources and supports, and make a meaningful impact on their first-year experience.

As an FYI mentor, you can:

- Develop your leadership, communication, and mentorship skills
- Build connections and relationships across campus
- Receive recognition on your Co-Curricular Record (CCR)
- Complete a 4-hour Brightspace training & receive an **\$80 ONECard top-up**

Eligibility:

- Have completed 12.0 undergraduate units within the last 5 years
- Be enrolled in both Fall 2026 and Spring 2027
- Be in good academic standing

Interested in becoming a mentor? Sign up here:

<https://uvic.mentorcollective.org/register/uvic-firstyear/mentor>



MULTI AGENT AI ORCHESTRATION

HANDS-ON

Asish George, Graduate Student, ECE

Build an AI that does things.
Not one that just answers.

YOU ASK

IT PLANS

IT ACTS

IT CHECKS

YOU APPROVE

What you build

- An agent that turns your goal into steps.
- An agent that works in the apps you already use.
- An agent that checks the work of the other two.
- One pipeline that runs all three.
- Nothing sends until you approve it.

The format

- 90 minutes, hands-on.
- Runs in the browser. Open source tools.
- Built for first-year to fourth-year.
- Leave with a multi-agent workflow you can reuse.

About the speaker

Seven years in industry as a software engineer and technical lead, building AI systems and cloud infrastructure. AWS certified. Publishes open source developer tools.

| Certificate of participation. Free food. 🍴

Friday 18 September
4:00 to 5:30 pm
ECS 660 Conference Room

Bring a laptop. No prior AI experience needed.



Register

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Outline: W02A

I) Data Mining (Review)

II) Course Logistics (Review)

III) Brightspace Tour

III) Another Brick in the Wall

Outline: W02B, Topics Start

Outline: W02A

I) Data Mining (Review)

II) Course Logistics (Review)

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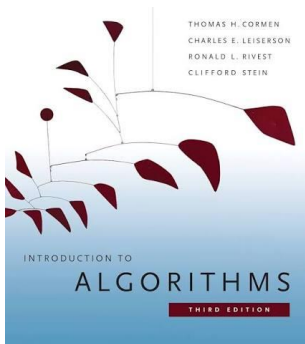
6

Background

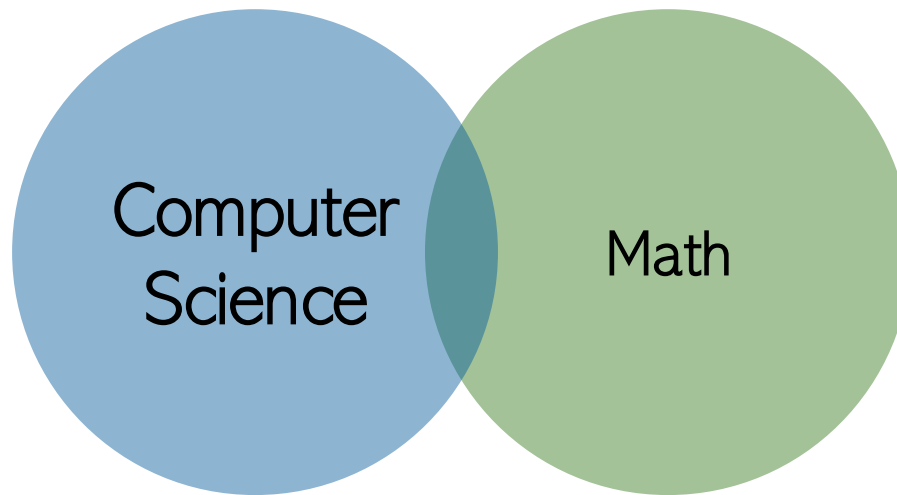
Design & Analysis of Algorithms

Greedy
Dynamic Programming
Divide-Conquer
Recursion
Backtracking
Big O

Data Structure & Modeling



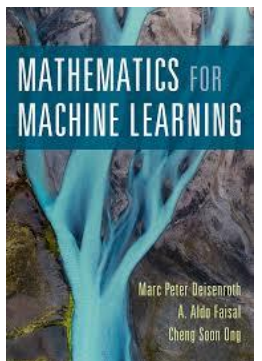
Introduction to Algorithms
By Cormen et al., 2009.



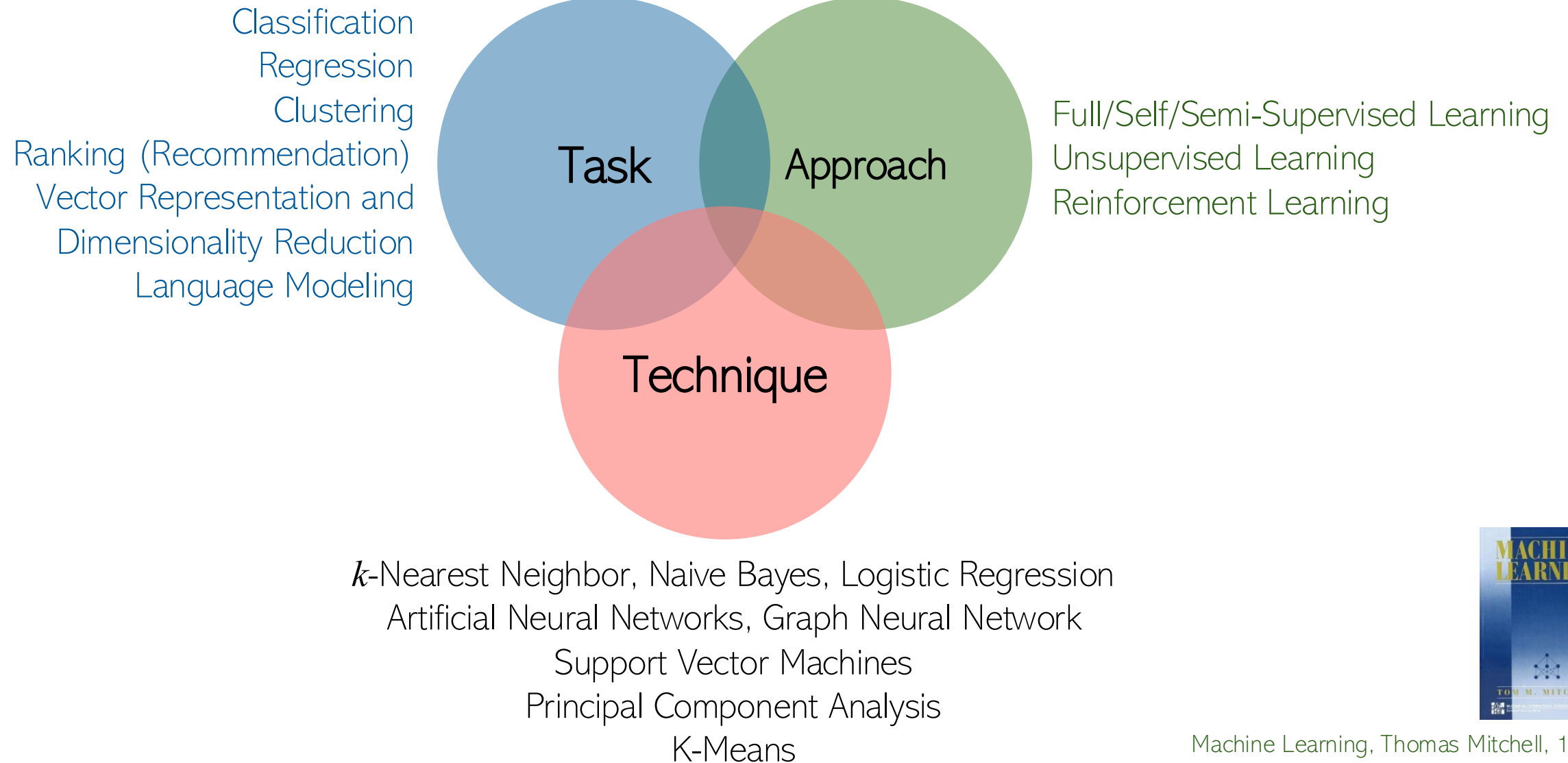
Multivariate Calculus

Derivatives
Partial Derivatives
Linear Algebra
Vectors
Matrices
Probability & Statistics

Mathematics for Machine Learning,
Deisenroth et al., 2020, <https://mml-book.github.io/>



Topics



Role

Data Mining Designer

Data Mining Developer

Data Mining Professional User

Data Mining End User

Who designs a new learning algorithm

Who applies existing learning algorithm

Who knows how learning algorithm works

Who uses products of learning algorithms

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Assessment

UG: Undergrad, GR: Grad

Team Project
UG: 50%
GR: 60%

Midterm Exam 1 (Week 07, Oct. 19)
10%

Midterm Exam (Week 10, Nov. 12)
10%

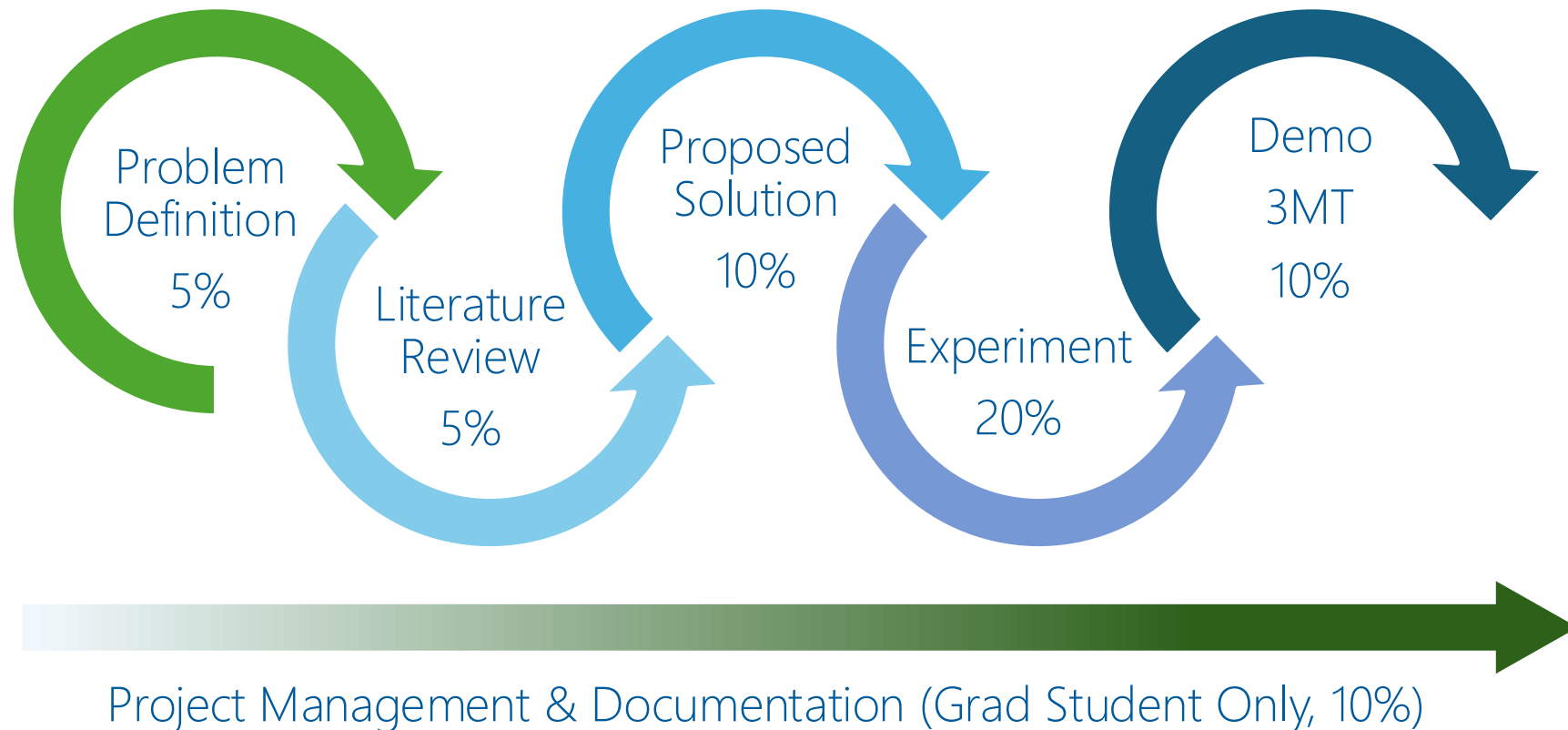
Final Exam (Week 15)
UG: 30%
GR: 20%

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Team Project

Min. 2, Max. 3 UG Members + 1 GR

Course-long Project vs. Final Project

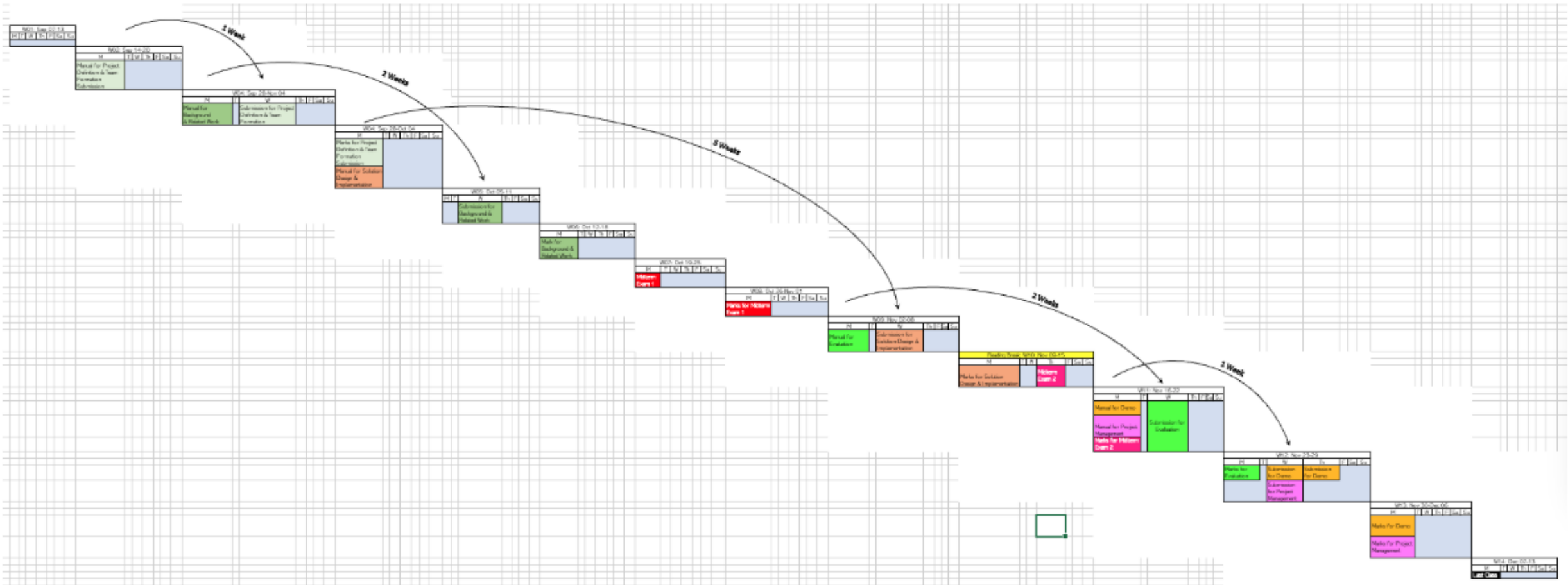


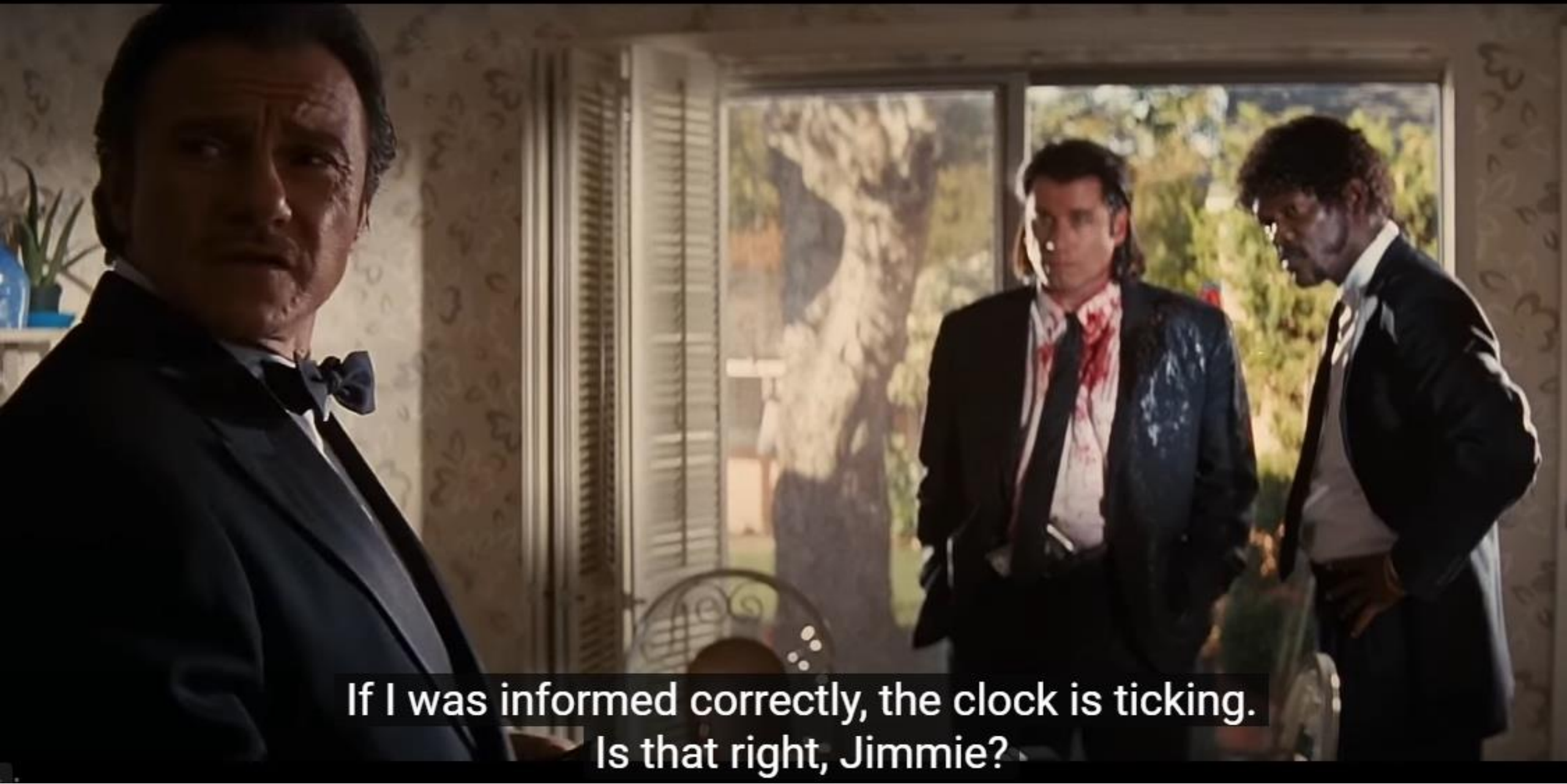
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Weekly Schedule

MON	TUE	WED	THU	FRI	SAT	SUN
New Submission Manual Prev. Submission Marks Release		Submission Deadline Midnight				
	B02: 10:30 - 11:20 Engineering Lab Wing B215: Reia	B03: 10:30 - 11:20 Engineering Lab Wing B215: Reia				
		B04: 11:30 - 12:20 Engineering Lab Wing B215: Ali				
		B05: 13:30 - 14:20 Engineering Lab Wing B215: Ali				
Lecture A: 2:30-3:30 Strong Building C103: Hossein		Lecture B: 2:30-3:30 Strong Building C103: Hossein	Lecture C: 2:30-3:30 Strong Building C103: Hossein			
Office A: 3:30-4:30 ECS613: Hossein		Office B: 3:30-4:30 ECS613: Hossein	Office C: 3:30-4:30 ECS613: Hossein			
		Marking				
Monitor Discussion Board						
Link Lecture Video		Link Lecture Video	Link Lecture Video			

Timeline





If I was informed correctly, the clock is ticking.
Is that right, Jimmie?

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Team Project: Take a Small but Novel Step!



Osprey: Enhancing Online Grooming Detection via Backtranslation Augmentation

<https://github.com/fani-lab/Osprey>

<https://doi.org/10.1145/3627673.3679974>

<https://aclanthology.org/2025.coling-main.160/>

Document Classification

Conversation Classification

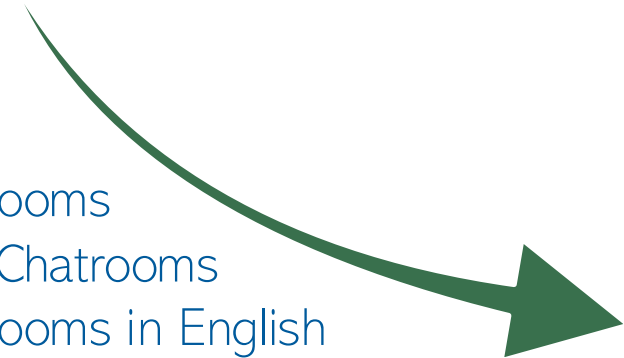
Conversation Classification in Chatrooms

Predatory Conversation Classification in Chatrooms

Predatory Conversation Classification in Chatrooms in English

The Effect of Augmentation on Predatory Conversation Classification in Chatrooms in English

The Effect of Backtranslation Augmentation on Predatory Conversation Classification in Chatrooms in English



16 Team Project: GitHub Repo

Branding: Pick a name for your project, design a logo, ...

Structured: Directories, Layers, Output, ...

Contributions: Pull Requests, Commits, ...

license.txt

readme.md: Explanation, setup, samples, scrips, ...

Let's have a look at some good and bad repos

https://github.com/hosseinfani/security_incident_ontology

<https://github.com/fani-lab/OpeNTF>

<https://github.com/fani-lab/LADy>

<https://github.com/fani-lab/Osprey>



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Assistants: For Help, Support, Sharing Experience, ...

Encouraged to form teams whose members are from the same lab

Reopening for another submission attempt

Reviewing markings of a submission

Accepting a late submission for a verifiable reason,

For **mixed team projects**, we'll assign a unique lab instructor.



Reia Drucker

reiadrucker@uvic.ca

Labs: B02, B03

Tue-Wed 10:30 AM-11:20 AM



Ali Mortazavi

alimorty@uvic.ca;

Labs: B04, B05

Wed 11:30 AM-12:20 PM, 01:30 PM-02:20 PM

More help ...

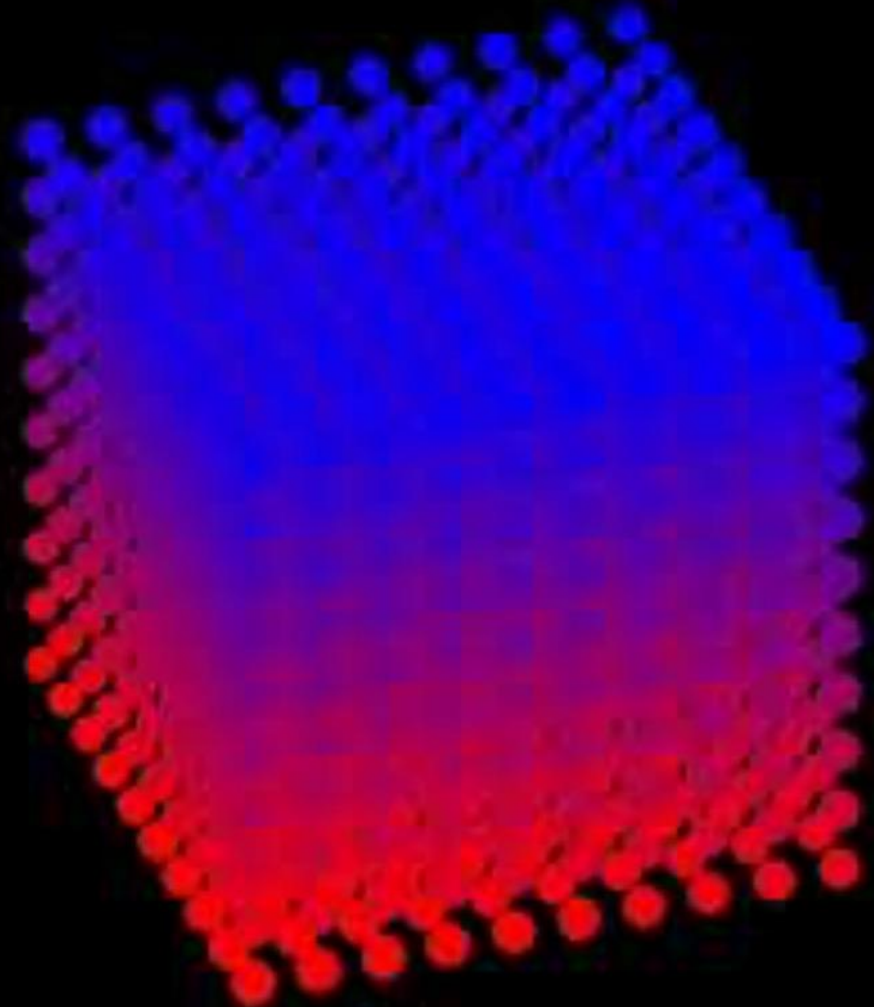


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Chios to Order: Be Patient

Shell Sort
100% Sorted (1 Segments)
15,494 Comparisons
Real Time: 35,41ms
0 Swaps
15,494 Writes to Main Array
0 Writes to Auxillary Array
1000 Elements

CompilerStuck



Outline: W02A

I) Data Mining (Review)

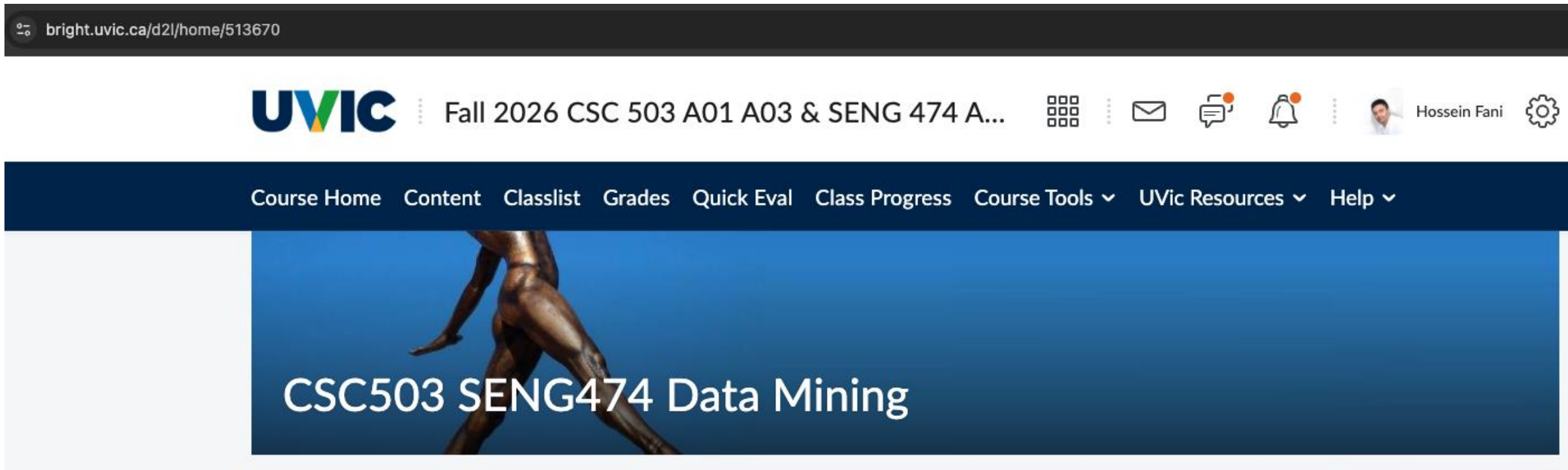
II) Course Logistics

III) Brightspace Tour

III) Another Brick in the Wall

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A Brightspace Tour



The screenshot shows the Brightspace interface for the course "Fall 2026 CSC 503 A01 A03 & SENG 474 A...". The URL in the address bar is `bright.uvic.ca/d2l/home/513670`. The UVic logo is on the left, and the instructor's name, Hossein Fani, is on the right. A navigation bar contains links for Course Home, Content, Classlist, Grades, Quick Eval, Class Progress, Course Tools, UVic Resources, and Help. Below the navigation bar is a large banner image of a bronze statue of a person in a dynamic pose, with the text "CSC503 SENG474 Data Mining" overlaid in white.

Outline

Instructor & Assistant

Schedule (Weekly, Timeline)

Lectures ([Forum](#), Slides, Recordings, Hyperlinks, Resources, ...)

Project ([Forum](#), Milestones, Manuals, ...) Milestone 1: Proposal: Problem Definition & Team Formation

Exams ([Forum](#), ...)

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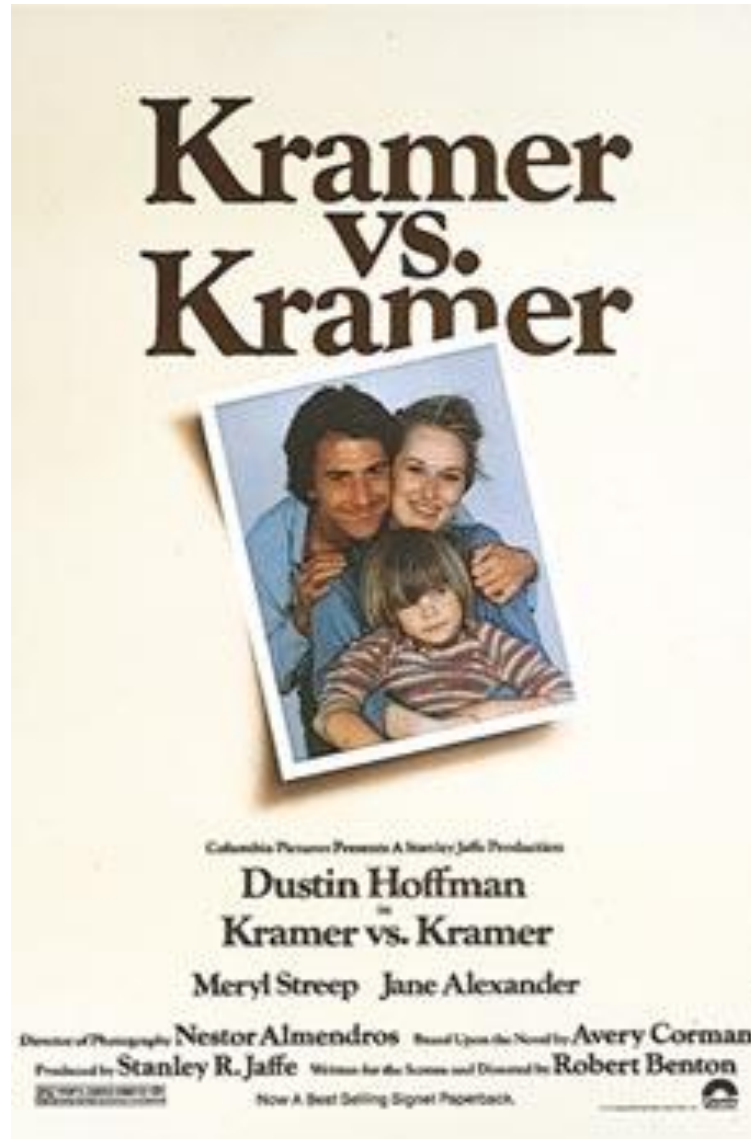
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Another Brick In The Wall, 1979, Pink Floyd

<https://www.youtube.com/watch?v=YR5ApYxkU-U>

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Googler vs. Googler



Googler: Who Builds Google (\$3-4 Trillion)

Googler: Who Works for Google (\$167K-655K)

Googler: Who Uses Google (\$60K-100K)



Larry Page and Sergey Brin in 2003

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A Feedback

QUALITY

2.0**COMP2650**

Mar 20th, 2025

For Credit: **Yes** Attendance: **Not Mandatory** Grade: **Incomplete** Textbook: **Yes**

Fani inspired me to not be another brick in the **wall**. Besides that, he cannot teach.

DIFFICULTY

4.0

INSPIRATIONAL

Helpful  0  0



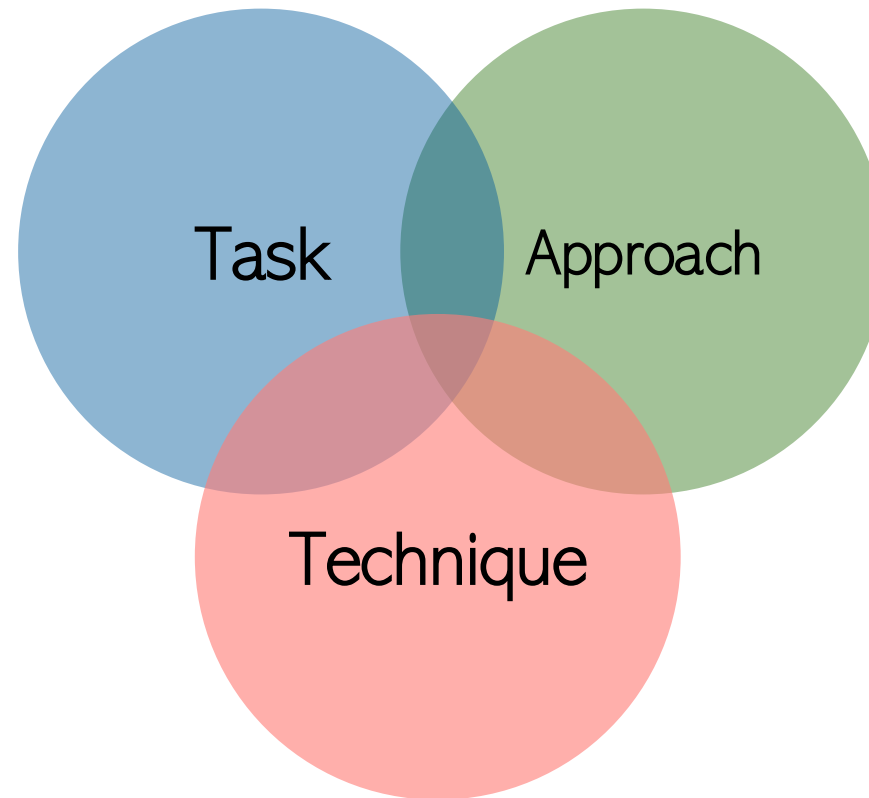
Take Aways

- Some great movies to watch ...
- Some wisdom on learning algorithms.
- Not to be another brick in the wall!

Outline: W02B: Topics Start

Classification

Regression
Clustering
Ranking (Recommendation)
Vector Representation and
Dimensionality Reduction
Language Modeling



Full/Self/Semi-Supervised Learning

Unsupervised Learning
Reinforcement Learning

k -Nearest Neighbor, Naive Bayes, Logistic Regression
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

Outline: W02B: Topics Start

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- III) Evaluation

Outline: W02B: Topics Start

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- III) Evaluation